Japan Airlines Flight 123

Flight model: Boeing 747
Date: August 12, 1985

From: Tokyo, Japan
To: Osaka, Japan

Boeing 747 is a jumbo jet.

Problem

12 mins into flight, Crew hear some explosion. The flight plunges up and down continuously. Air rushes into the cabin. There is a sudden loss of pressure which means there must be a hole in the aircraft. All the hydraulics is dropping. The hydraulics are considered to be the lifeline of a plane. The crew starts experiencing **hypoxcia**, loss of oxygen to the brain. Which make their judgment impaired. In one of the rows, the oxygen supply has failed. The crew says that they also need to wear a oxygen mask but none of them wears it. The pilots find that when the plane is diving, increasing the engine power pulls the nose up. And when the plane is climbing deceasing the power pulls the noses drops. Also, by applying power to the left they are able to turn the plane to the right and vice-verse. To decrease the speed, the crew lowers the landing gear. This creates a drag slowing the plane down but the plane losses directional control. Near mount fuji, the plane make a abrupt turn to the right and dives. Then, the plane again climbs up. Then the plane makes left and crashes into the mountain

Investigation

In one of the photograph captured by a photographer it was clear that the fin part of the plane was missing. All the hydraulics passes through this which would have lead to loss of hydraulics. Checking the history it was found that 7 years earlier, the plane had faced a tail strike. There was a repair to maintain the pressure. When the 747 on land it looks like oval shape and when on air it looks more circular, this is due to change in air pressure. The **AFT pressure bulk head** in the tail of the plane is like a huge umbrella this prevents pressurized air escaping the cabin. Boeing was called to repair the AFT bulk head after the tail strike. Boeing did remove few panels and replaced with new panels but had added less reverts to secure the panel. The up and down movement is because when the plane falls it gains speed and lift and then it moves up then it looses speed to fall again this kept on repeating.

Changes that were brought after the incident

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Reference

1. The Deadliest Plane Crash Ever — Mayday Air Disaster
   `https://www.youtube.com/watch?v=SjhyWD9w-Yo`